S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 9

Aim: Evaluate Bayes Rules for Rain predication from given scenario.

Name of Student:

Roll No.:

Semester/Year: V/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Evaluate Bayes Rules for Rain predication from given scenario.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of Bayes theorem.
- To be able to solve problem based on Bayes theorem.

THEORY:

Bayes Theorem: - Bayes' Theorem states that the conditional probability of an event, based on the occurrence of another event, is equal to the likelihood of the second event given the first event multiplied by the probability of the first event.

Bayes theorem is used to find the reverse probabilities if we know the conditional probability of an event.

Bayes' Theorem states the following for any two events A and B:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

where:

· P(A|B): The probability of event A, given event B has occurred.

· P(B|A): The probability of event B, given event A has occurred.

· P(A): The probability of event A.

· P(B): The probability of event B.

Terms Related to Bayes' Theorem

After learning about Bayes theorem in detail, let us understand some important terms related to the concepts we covered in the formula and derivation.

- **Hypotheses:** Hypotheses refer to possible events or outcomes in the sample space; they are denoted as E1, E2, …, En.

Each hypothesis represents a distinct scenario that could explain an observed event.

- **Priori Probability:** Priori Probability P(Ei) is the initial probability of an event occurring before any new data is taken into account.

  It reflects existing knowledge or assumptions about the event.

Example: The probability of a person having a disease before taking a test.

- **Posterior Probability:** Posterior probability (P(Ei|A) is the updated probability of an event after considering new information.

It is derived using the Bayes Theorem.

Example: The probability of having a disease given a positive test result.

- **Conditional Probability:** The probability of an event A based on the occurrence of another event B is termed conditional Probability.

  It is denoted as P(A|B) and represents the probability of A when event B has already happened.

- **Joint Probability:** When the probability of two or more events occurring together and at the same time is measured, it is marked as Joint Probability.

  For two events A and B, it is denoted by joint probability is denoted as $P(A \cap B)$.

- **Random Variables:**

  Real-valued variables whose possible values are determined by random experiments are called random variables.

  The probability of finding such variables is the experimental probability.

Rain Prediction Example: Suppose the probability of the weather being cloudy is 40%. Also suppose the probability of rain on a given day is 20%.

Also suppose the probability of clouds on a rainy day is 85%.

If it's cloudy outside on a given day, what is the probability that it will rain that day?

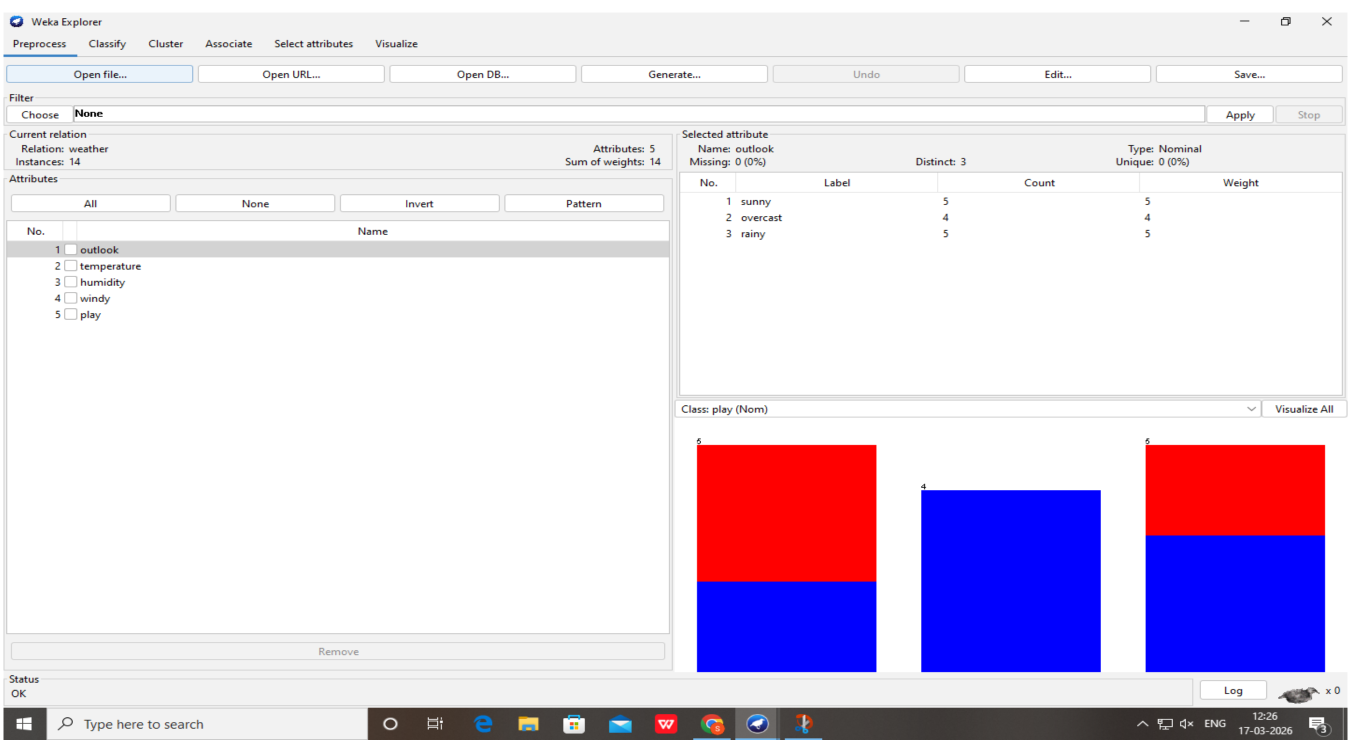
Solution:

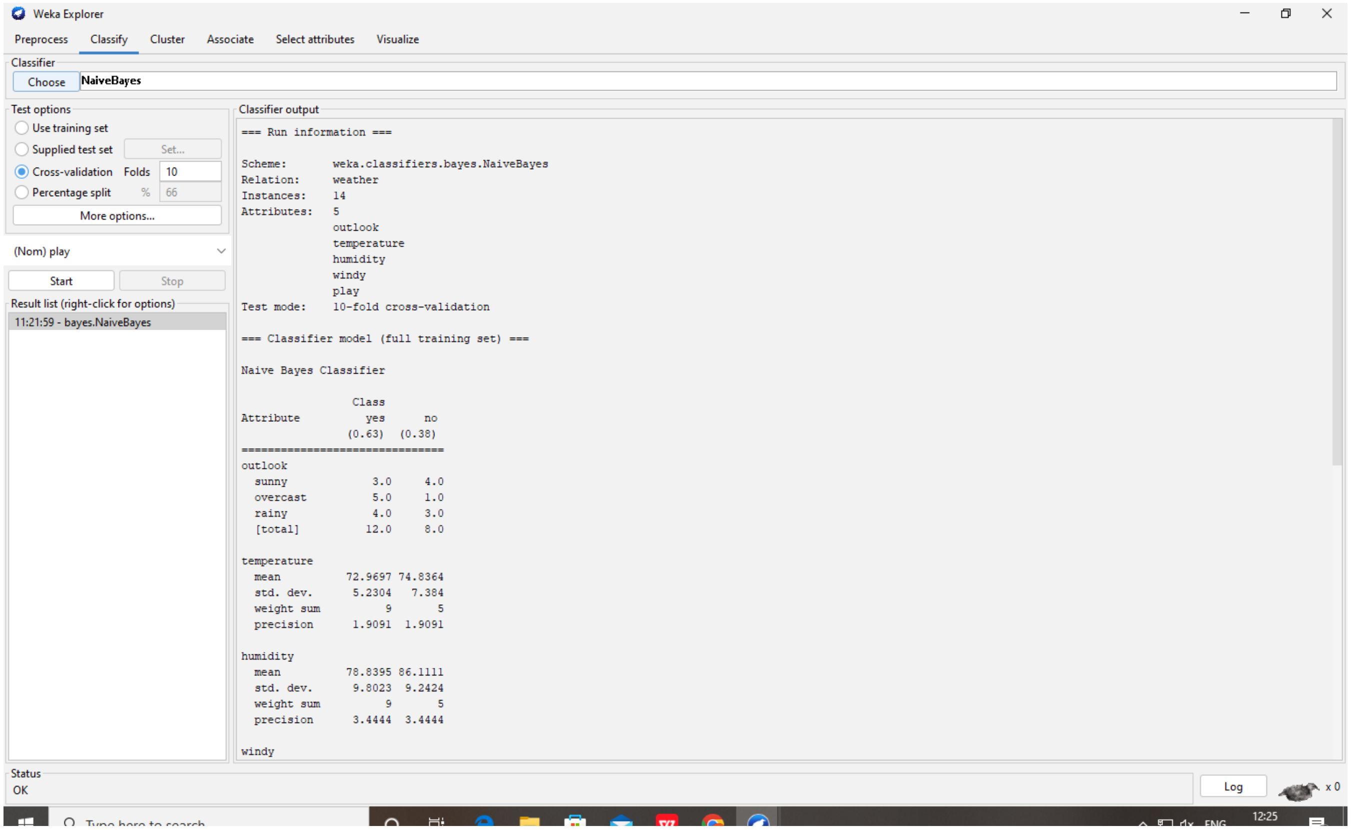
· P(cloudy) = 0.40
· P(rain) = 0.20
· P (cloudy | rain) = 0.85
· Thus, we can calculate:
· (rain | cloudy) = P(rain) * P (cloudy | rain) / P(cloudy)
· P (rain | cloudy) = 0.20 * 0.85 / 0.40
· P (rain | cloudy) = 0.425
· If it's cloudy outside on a given day, the probability that it will rain that day is 42.5%.

PROGRAM CODE:

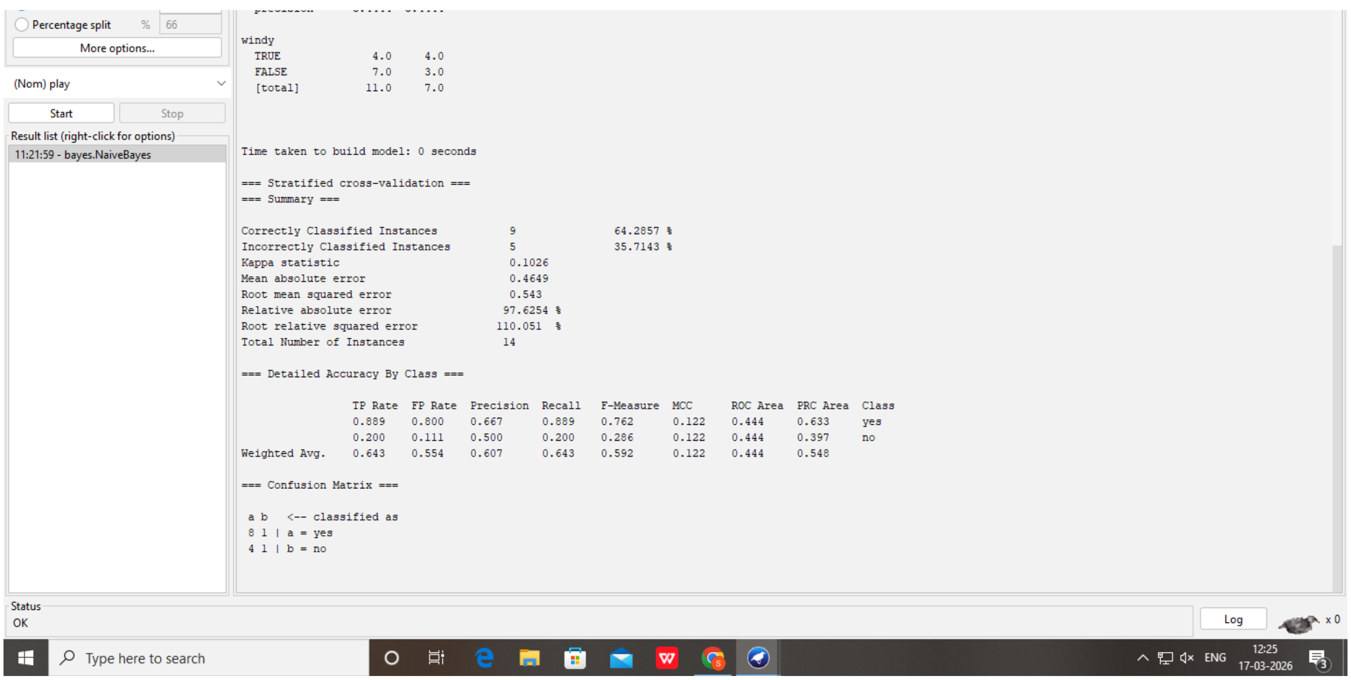
```
@relation weather

@attribute outlook {sunny, overcast, rainy}
@attribute temperature numeric
@attribute humidity numeric
@attribute windy {TRUE, FALSE}
@attribute play {yes, no}

@data
sunny,85,85,FALSE,no
sunny,80,90,TRUE,no
overcast,83,86,FALSE,yes
rainy,70,96,FALSE,yes
rainy,68,80,FALSE,yes
rainy,65,70,TRUE,no
overcast,64,65,TRUE,yes
sunny,72,95,FALSE,no
sunny,69,70,FALSE,yes
rainy,75,80,FALSE,yes
sunny,75,70,TRUE,yes
overcast,72,90,TRUE,yes
overcast,81,75,FALSE,yes
rainy,71,91,TRUE,no
```

INPUT & OUTPUT:






```
=== Run information ===

Scheme:       weka.classifiers.bayes.NaiveBayes
Relation:     weather
Instances:    14
Attributes:   5
              outlook
              temperature
              humidity
              windy
              play
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

                 Class
Attribute          yes      no
                (0.63)  (0.38)
===============================
outlook
  sunny            3.0     4.0
  overcast         5.0     1.0
  rainy            4.0     3.0
  [total]         12.0     8.0

temperature
  mean         72.9697 74.8364
  std. dev.     5.2304   7.384
  weight sum         9       5
  precision     1.9091  1.9091

humidity
  mean         78.8395 86.1111
  std. dev.     9.8023  9.2424
  weight sum         9       5
  precision     3.4444  3.4444

windy
```




```
windy
  TRUE             4.0     4.0
  FALSE            7.0     3.0
  [total]         11.0     7.0



Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances           9               64.2857 %
Incorrectly Classified Instances         5               35.7143 %
Kappa statistic                          0.1026
Mean absolute error                      0.4649
Root mean squared error                  0.543
Relative absolute error                 97.6254 %
Root relative squared error            110.051  %
Total Number of Instances               14

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.889    0.800    0.667      0.889    0.762      0.122    0.444     0.633     yes
                 0.200    0.111    0.500      0.200    0.286      0.122    0.444     0.397     no
Weighted Avg.    0.643    0.554    0.607      0.643    0.592      0.122    0.444     0.548

=== Confusion Matrix ===

 a b   <-- classified as
 8 1 | a = yes
 4 1 | b = no
```

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is the correct formula for Bayes Theorem?

2. Which algorithm uses Bayes theorem?

3. When can we use Bayes theorem?

4. How is Bayes theorem used in real life?

REFERENCES:

- *https://www.geeksforgeeks.org/maths/bayes-theorem/*
- *https://discovery.cs.illinois.edu/learn/Prediction-and-Probability/Bayes-Theorem/*
- *https://blogs.cornell.edu/info2040/2021/11/05/sunny-with-a-chance-of-rain-using-bayes-theorem-to-predict-the-weather/*